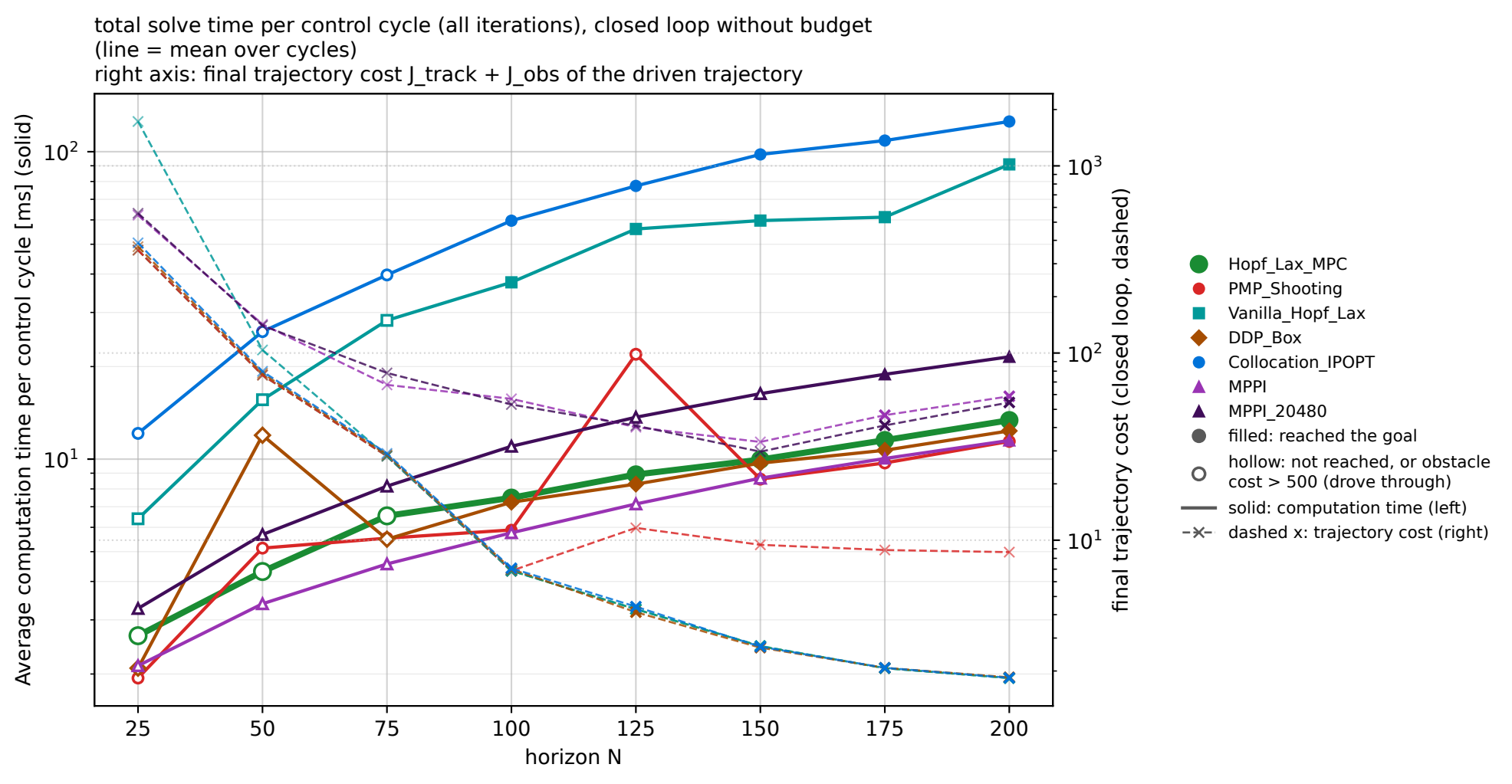

total solve time per control cycle (all iterations), closed loop without budget
(line = mean over cycles)
right axis: final trajectory cost J_track + J_obs of the driven trajectory
Average computation time per control cycle [ms] (solid)
final trajectory cost (closed loop, dashed)
horizon N
10^2
10^1
10^3
10^2
10^1
25
50
75
100
125
150
175
200
Hopf_Lax_MPC
PMP_Shooting
Vanilla_Hopf_Lax
DDP_Box
Collocation_IPOPT
MPPI
MPPI_20480
filled: reached the goal
hollow: not reached, or obstacle cost > 500 (drove through)
solid: computation time (left)
dashed x: trajectory cost (right)